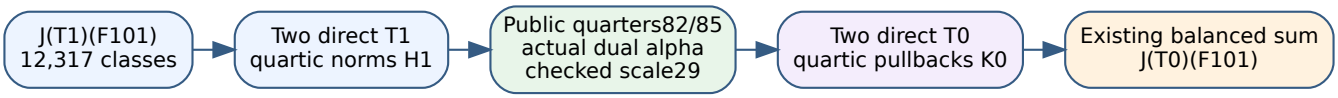
The actual first Jacobian adjoint using F101 arithmetic

2026-11-12 · T1 to T0 · scientific standing unsigned

The actual SevenChain.dual now selects a standalone prime-field adjoint. Complete changed check.sh03 exits0:25native/92reference unit+five integrations,one pre-existing ignored benchmark,all original source guards, fmt and strictClippy. PCCd744cf pins the whole136317-byte raw log as standardXZ15244bytes,raw/archive hashes,exact roundtrip and all nine unchanged source02 hashes after the run. The exact original dual body remains dual_reference and the new full comparison calls it explicitly. [pcc:unverified]



Existing engines and exact reverse maps

The evaluator takes both direct T1 norm coordinates,applies public quarters82/85,evaluates the selected actual elliptic duals,then uses two T0 quartic pullbacks and the existing balanced addition. It reuses existing field,polynomial,elliptic,Kohel and Balanced engines. The standalone constructor/evaluator has no Richelot or extension-field stage;the retained comparison context still constructs old maps for reference controls. [pcc:unverified]

E109 prime=(26,58)	E113 prime=(69,50)
reversekernel=[58,26,58,1]	reversekernel=[29,88,45,1]
normalized codomain=(78,72)	normalized codomain=(99,68)
scale29 returns(1,46)	scale29 returns(44,21)

The two short-Weierstrass coefficient equations check the scale29=1/7 in F101. Selected domains,codomains,kernels,numerators and denominators match the retained maps exactly. The complete test checks every12,317 olddual output,bijection,all fixed-partner sums,negations,identity and wrong-model inputs. Both forward-adjoint compositions equal multiplication by7 on every class in both rational domains. Every109/113 reverse embedded elliptic-source endpoint is also checked. [pcc:unverified]

The T1 pole cases remain in the norm domain

Each direct norm matches every12,317 oldH1 output and all negations, fixed-partner sums,infinities and model controls. Per component,the test includes4linear-pole inputs,232partial-pole quadratics,2all-pole supports and118repeated supports. Complete route counts are [3,236,2,232,110,106,11628] and[3,236,2,232,132,106,11606]. These are retained cases of the existing finite-algebra adapter. T1norm18 and the adjoint24 constructor/evaluator arithmetic have exact source custody. [pcc:unverified]

Why this computes the accepted adjoint

The accepted global forward factorization is4Psi=K1 alpha H0,where H_i are the pair of degree4 norms and K_i their Picard adjoints. Taking adjoints gives4PsiHat=K0 alphaHat H1. The elliptic rational groups have orders109 and113,with4 times82=1mod109 and4 times85=1mod113. The destination Jacobian rational group has odd order12,317,so multiplication by4 is injective there. The displayed rational-point algorithm therefore computes the old PsiHat. It supplies no global inverse morphism of[4]. PRIME_FIELD_ADJOINT.md retains upstream526f9f/4934b4/78176c and the old quotient,Picard,polarization and norm dependencies. [pcc:unverified]

For a finite elliptic point P=(X,Y),the pullback uses the whole fibre U=N-XQ² and V=YQ³/M moduloU. On this finite algebra the cover/target identities giveF M²=Y²Q⁶,so Q and M are units even for repeated support. The finite fibre and actual infinity multiplicities enter the unchanged balanced reduction. The odd rational elliptic groups exclude nonzero two-torsion;its geometric chart remains outside this implemented domain. The original finite-algebra proof and bound QuarticFibre helpers remain explicit dependencies;geometric formalization is partial. [pcc:unverified]

Source custody and failure preservation

Sourceguard02/ccfb25 checks that the original forward,forward_reference, quarter and naive bodies and the old test suffix remain unchanged. The old dual is retained byte-for-byte as dual_reference. Existing Balanced, quartic cover,Lagrange and point sources are unchanged. The forward key helper has a visibility-only change;the point helper delegates to the already loaded enumerator. No second group or field engine was added. All original90 tests remain;the two added tests account for92. [pcc:unverified]

Initial integration01 failed compilation because a moved test accessed a private forward-context field. The exact failure/source bb323e is preserved. The test now calls the existing Pullback::fixed(1,i) API; constructor/evaluator arithmetic and all acceptance criteria are unchanged. The producer prototype’s earlier unused-helper failure and prior forward interruption/cache/field-assumption records remain upstream. [pcc:unverified]

Parent, integration and next scope

This ready child startsb2972eb51fd794e1a6061c41fd5f3828890c08e3 and targets immediate1506. Parent exact focused43180569278 succeeds25native/ 90reference+five;complete decoded220108-byte log→XZ28636/API/hash/ roundtrip64e1d9 is archived here. The new92-test head requires its own focusedCI/review. Integrate bottom-up after actual gates,retarget/rerun after parents land. Frozen-ledger forks and parent1484 C/conflict findings remain;no gate or lineage bypass is implied. [pcc:unverified]

Separate upstream T2/compressed-chain results8e0523/4ac90f and geometric argumentb05286 are outside this actual92-test checkpoint. They are the next actual integration. Existing root138written helpers/26modules remain partial structural formalization. Matched cold construction,transfer, solve,recovery and verification costs await S3/livebudget admission; no timing or solver advantage is claimed. KB/ecdb unavailable;atlas vault locked. [pcc:unverified]